

LAB 14: BALMER SERIES AND RYDBERG CONSTANT

During this laboratory session, students will

- ✓1 operate hydrogen discharge tube and diffraction grating,
- ✓2 observe and quantify interference patterns,
- ✓3 measure different wavelengths of the hydrogen spectrum using a diffraction grating,
- ✓4 consult the National Institute of Standards and Technology (NIST) data online,
- ✓5 link wavelength to energies,
- ✓6 measure the so-called Rydberg constant,
- ✓7 have their first laboratory that touches quantum mechanics.

Student Learning Outcome: Successful students will



- ✓1 master the so-called Balmer series for hydrogen,
- ✓2 be aware of the existence of other series that are not in the visible range,
- ✓3 be acquainted to electronic states and radiative decay in general.

INTRODUCTION

The light emitted by hydrogen atoms when heated does not form a continuous spectrum as you will see in this lab. The emission is quantized. J. J. Balmer derived an empirical formula that described the data accurately in 1885 though his consideration was without theoretical basis. His equation predicts the wavelength of the emission line.

$$1/\lambda = R (1/2^2 - 1/n^2) \quad (1)$$

where $n = 3, 4, 5, \dots$ (whole number called the principle quantum number later on),

λ = the wavelength of the emission line,

R = the Rydberg constant that you will have to measure during this laboratory. The expected value of R is $1.097 \times 10^7 \text{ m}^{-1}$.

Your goal is to verify Balmer's empirical formula and to measure the Rydberg constant. You will have full agency for this laboratory. A hydrogen lamp discharge and a diffraction grating (around 1,000 lines/mm) are available to you to conduct the experiment. You should be able to see at least three main spectral lines of the hydrogen atom (red, blue-green, and violet) that you will use to measure the Rydberg constant. Each wavelength could be computed by using the following diffraction grating equation after the measurement of the angle θ .

$$m * \lambda = d * \sin \theta \quad (2)$$

You may look back to one of the previous laboratories entitled “Understanding diffraction grating” when creating your laboratory procedure.

Form Availability Start Date:

Mar 21, 2026 @ 07:00 AM

Form Availability End Date:

Mar 23, 2026 @ 02:00 PM

LAB 14: PRELIMINARY WORK

Go to **Strong Lines of Hydrogen (H)** (<https://physics.nist.gov/PhysRefData/Handbook/Tables/hydrogentable2.htm>), which is a webpage of the National Institute of Standards and Technology (NIST). According to the site, what are the strongest lines for H (I) that are in the visible range?

Using equation (1) in the Introduction section, find the values of “n” for each wavelength you listed in #1.

Assume that your diffraction grating has 1,000 lines/mm and it is 1 m away. Calculate the angles at which each line diffracts for the first-order spectrum (that is to say $m = 1$).

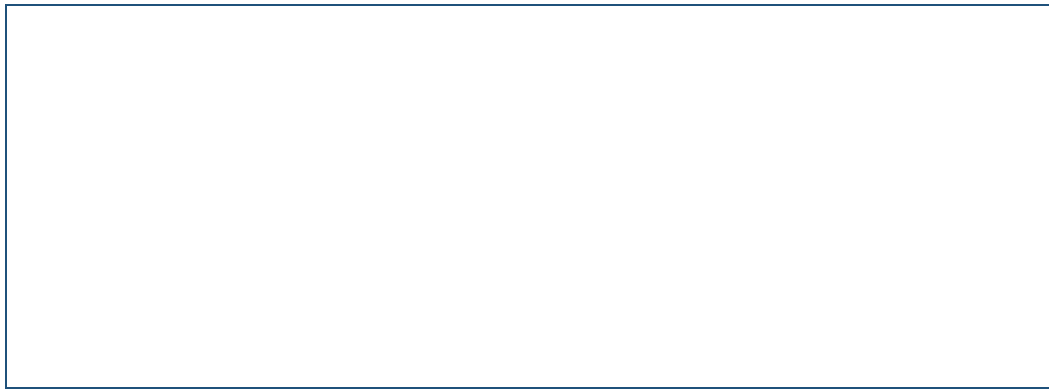
What would be the possible values of “m” (orders) in equation (2) for each of the spectrum lines?

If we replace $1/2^2$ of equation (1) with $1/3^2$, what would be the first expected wavelength?

Is the wavelength you calculated in #5 part of the visible range?



Which one of the red, blue-green, and violet lines corresponds to $n = 3$? Why?



Activity #1: Measuring the number of lines per unit length of the diffraction grating

You are given a red laser to measure the number of lines per unit length of a grating system. Include a description of how you would proceed (drawing and formula with explanation) in the laboratory report.

Activity #2: Verify Balmer's formula experimentally and find R

Do not forget that you will have to use the experimental number of lines per unit length measured in activity #1. Also, explain your method in your laboratory report. The following image shows the experimental set-up.



Source: Vola Andrianarijaona



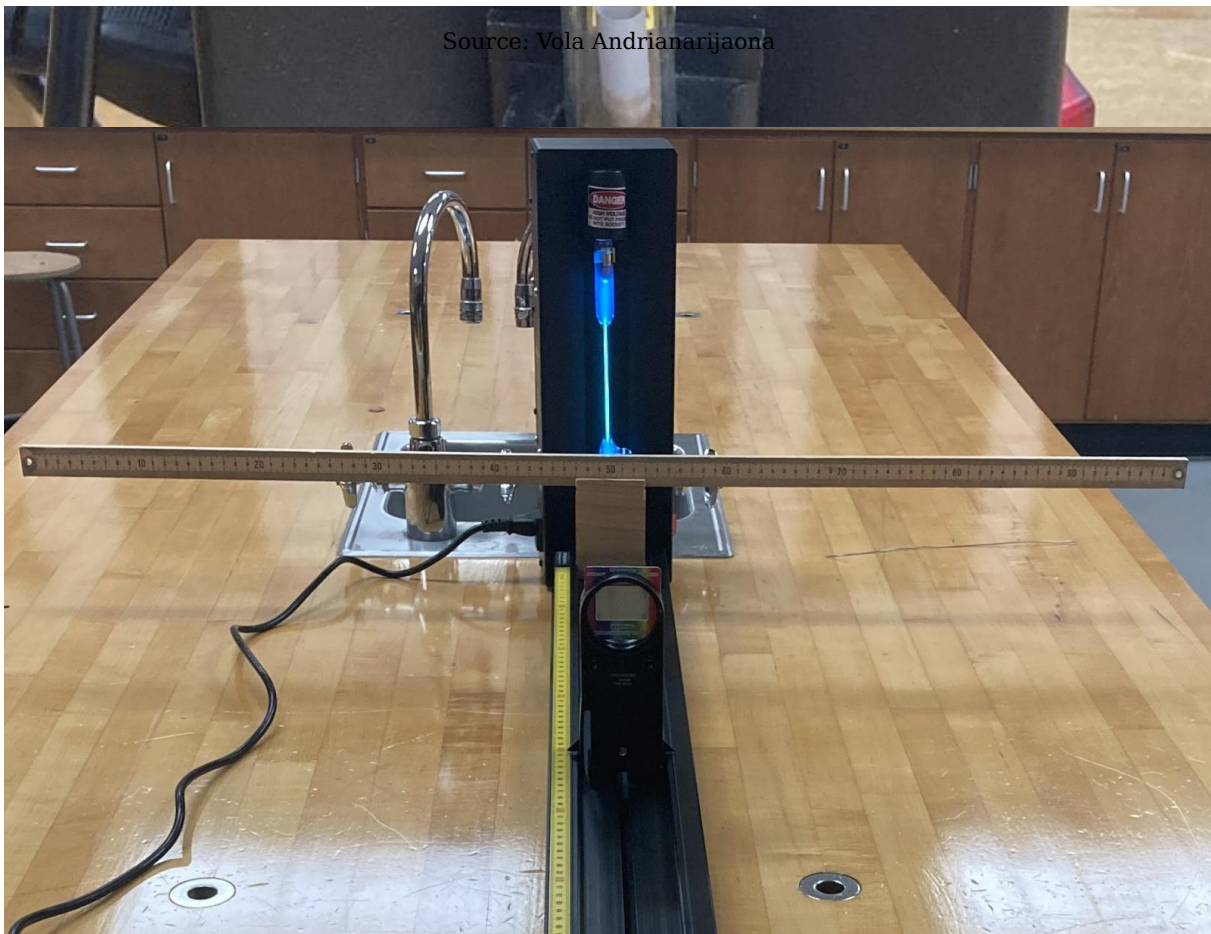
First, insert the hydrogen lamp discharge into the lower nozzle of the power supply. Make sure you feel the reaction of the spring of the lower nozzle to secure the electric contact. Call the instructor if you are not sure whether the lamp is in its correct place in the lower nozzle or not.

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HIGH VOLTAGE
DO NOT PUT FINGERS
INTO SOCKETS

ELECTRIC



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